

Desktop Automation Capability Specification

Project NOVA Desktop Orchestration, Window Management, and OS Automation Interface

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Revision History

Version	Date	Author	Summary of Changes
1.0	2026-06-28	Antigravity	Consolidate Desktop Overview, App Management, Window Control, Mouse/Keyboard Emulation, Screen Capture, Safety, and

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Purpose & Scope

This specification defines the functional boundaries, API contracts, and safety policies for the **Desktop Automation Capability**. It permits NOVA to open applications, position windows, capture displays, and emulate mouse and keyboard inputs under standard operating system environments.

Application Management

Functions

- **Open Application:** Launches a target program at an absolute path with configurable string arguments.
- **Close Application:** Safely terminates a process given its process identifier (PID).
- **Focus Application:** Moves a target process window to the foreground.
- **Detect Running Processes:** Queries the OS process table to return running instances.

Acceptance Criteria

Applications launch, switch focus, and close reliably, returning structured result objects containing process metadata.

Window Management

The subsystem must wrap window handle operations to manage application displays:

- **Enumerate Windows:** Returns a list of active application handles, titles, and positions.
- **Window Control:** Minimizes, maximizes, restores, or resizes window states.
- **Window Positioning:** Moves window bounds to target display screen coordinates.

Mouse Emulation Control

Provides mouse cursor inputs:

- **Move Cursor:** Glides mouse coordinates to a target point.
- **Emulate Clicks:** Performs left clicks, right clicks, double clicks, or drags.
- **Scroll Wheel:** Emulates mouse wheel rolls (vertical and horizontal).

Safety constraint: Mouse inputs must validate coordinates against physical monitor bounds. Off-screen emulations are prohibited.

Keyboard Emulation Control

Provides keyboard input automation:

- **Type Text:** Emulates key strokes for a target string block.
- **Key Combinations:** Emulates concurrent key actions (e.g. `Ctrl+C`).
- **Hotkeys:** Triggers system global hotkeys.
- **Clipboard Transfers:** Reads and writes text values to the OS clipboard.

Display Capture & Screen Observation

Captures screen pixels for vision processing:

- **Full Screenshot:** Acquires a PNG buffer of the primary screen.
- **Region Capture:** Acquires a specific coordinate bounding box.
- **Active Window Frame:** Captures the focused application window boundary.
- **Multi-Monitor Support:** Detects multiple connected displays and handles offset coordinates.

Constraint: Screen Capture only outputs raw image buffers; it does not parse layouts or execute character recognition (OCR) natively.

Safety, Permissions & Kill Switches

To guarantee safety, the implementation enforces these non-negotiable checks:

9. **Permission Validation:** All desktop actions check permission grants with the ``PermissionEngine`` before execution.
10. **Emergency Abort Hotkey:** A background global listener checks for a dedicated cancel hotkey sequence (``Ctrl+Alt+Shift+X``). Triggering this immediately cancels active cursor moves or keyboard inputs and halts execution loops.
11. **Audit Logs:** Every click, keystroke, process launch, and abort is recorded in the local Audit Log.
12. **Allow/Deny Filters:** Process directories and command targets are restricted using configurable regex filter files.

Testing & Verification Suite

- **Unit Tests:** Every module tests process launches and window queries using mock OS API structures.
- **DPI Scaling Tests:** Verification runners check mouse clicks calibrate correctly under differing OS DPI scaling offsets (e.g. 150% scaling).
- **Failure Recovery Gaps:** Tests verify the execution engine aborts gracefully if a target application window hangs or crashes.
- **Multi-Monitor Checks:** Confirms coordinate conversions are safe on secondary display configurations.